Ryan Nguyen

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Education

University of California, Berkeley **Expected Graduation: December 2027**

Bachelor's in Computer Science

- **Scholarship Foundation Exceptional Leadership Scholarship** Recipient
- Relevant Coursework: Data Structures & Algorithms, Efficient Algorithms & Intractable Problems, Computer Architecture, Operating Systems, Machine Learning, Databases, Discrete Mathematics & Probability
- Campus Involvement: **Web Development @ Berkeley** (President of Boot-camp Development & Industry Developer)

Experience

Software Development Engineering Intern Aug 2026 - Present

Amazon Web Services *Seattle, WA*

- Engineered an end-to-end SMT solver integration pipeline for Amazon Trusted Solver (ATS), **reducing release adoption time from 5 days to 4 hours** by **automating 95% of manual engineering work** across the release lifecycle.
- Designed a distributed release orchestration system using **Lambda, CodeBuild, and DynamoDB conditional writes**, ensuring atomic release claiming, concurrency-safe execution, and automated recovery across solver integration workflows.
- Developed a deterministic regression evaluation engine leveraging PAR-2 scoring, per-bundle performance analysis, and baseline-relative soundness checks, enforcing 4 fail-closed safety invariants with **zero unsafe approvals across 198 tests**.
- Integrated Amazon Bedrock LLMs into a safety-gated release decision engine, **automating 78% of onboarding decisions with 98.5% agreement** against scientist-reviewed benchmarks, validating recommendations against 42 release documents.

Software Engineering Intern Jun 2026 - Aug 2026

Genuine Parts Company *Atlanta, GA*

- Built a 360° customer intelligence platform and **Elasticsearch** service integrating cross-domain data aggregation.
- Rearchitected the data retrieval pipeline, parallelizing **14+ independent BigQuery workloads** with multithreading and asynchronous execution, reducing end-to-end search latency by **86% (23s to 3s)**.
- Designed multi-tier Redis caching layer, accelerating lookups from **3s→100ms**, reducing BigQuery compute costs by **80%**.
- Optimized frontend performance through concurrent data fetching, progressive rendering, hover prefetching, and compressed API responses, reducing time-to-first-content by **33% (3s to 2s)** and **eliminating blocking UI behavior**.

Technical Lead Jan 2026 - May 2026

LearnHaus AI (Techstars-Backed) *Granite Bay, CA*

- Led **9 engineers** to build and launch the company's first **LLM evaluation platform**, standardizing benchmarking across **4+ frontier models** and reducing model evaluation cycles by **95% (2–3 days to under 2 hours)**.
- Engineered **evaluation and regression analysis tooling** (MAE, variance, correlation, clustering), identifying **12+ scoring biases across 1K+ production outputs** through scoring, human-model comparisons, and dataset-level analysis.

Teaching Assistant, Full-Stack Development [Link] Nov 2025 - May 2026

- Engineered course infrastructure supporting **100+ students per semester**, developing **20+ pytest/Jest autograders** for full-stack applications, a submission platform handling **2,000+ assignments**, and **15+ automation scripts.**

Projects

Volare — AI Travel Platform [Link]| *React, TypeScript, Spring Boot, WebFlux, Resilience4j, Docker*

- Built and deployed a **full-stack AI travel platform** for daily itineraries, real-time flight search, and restaurant discovery.
- Engineered a reactive backend with **WebFlux, Resilience4j circuit breakers/retries, and async caching**, reducing cache-hit latency by **99% (1ms vs. 400–600ms upstream)**, **95% less redundant API calls** during upstream failures.

LockIn [Link]| *JavaScript, Chrome MV3, MutationObserver, CSS* `:has()`

- Engineered a lightweight **5.5KB Chrome extension** with **zero dependencies** that removes YouTube Shorts and Instagram Reels across **5 layout variants** (including A/B tests) using a **debounced MutationObserver**.
- Pivoted from CSS `:has()` to a JS attribute-stamping hide system after profiling render jank on YouTube's virtualized feed; **7 hand-tuned selectors** target every known Shorts and Reels DOM variant with **zero false positives**.

HTTP Reverse Proxy + Load Balancer [Link]| *Go,* `net/http/httputil`

- Architected a **zero-dependency HTTP reverse proxy and load balancer** in Go, implementing **3 load-balancing algorithms** (round-robin, least-connections, IP-hash) with per-route backend pools and longest-prefix routing.
- Engineered fault tolerance via health-checked backend eviction/re-admission and **bounded retry-on-5xx across up to 3 backend replicas**, enabling seamless failover and masking upstream failures for idempotent requests.

Technical Skills

Languages: Python, Java, Go, C/C++, JavaScript, TypeScript, SQL, RISC-V Assembly
Frameworks & Databases: React, Next.js, Express.js, Spring Boot, FastAPI, PostgreSQL, Redis, Elasticsearch, DynamoDB
Cloud & Infrastructure: AWS (Lambda, CodeBuild, EventBridge, Bedrock, S3, IAM), GCP (BigQuery, Vertex AI), Docker
AI & Systems: PyTorch, TensorFlow, LLM Evaluation, SMT Solvers (Z3, cvc5), REST APIs, CI/CD, Distributed Systems